

Background

In Cobre we have a transactional, cloud-native, event driven and microservices platform that manages resources available for the clients, such as, accounts, payments or transactions, enabling the capabilities to create, delete or update each one of them.

The Challenge.

Delivery of event notifications.

The Cobre Product team has defined a new feature in the platform-generated to notify individually about the different events generated by the platform (i.e. balance update or event creation) via a webhook to a specific URL. The notification feature must have the following capabilities:

- Confirm with a subscription if such an event has to be delivered. It's mandatory to ensure notifications sent to every client belong to events generated to that client.
- Delivery the notification to the specified URL endpoint.
- Handling error with an efficient retry strategy.
- Storing the final information related to the delivery.
- The capability must be observable in a near real-time approach to allow our internal monitoring team to promptly identify behavior deviation and promptly respond to any client complaint about the notification delivery.

Event notification self service API

Additionally, Cobre Product team promotes a self-service approach to our clients and they want to offer the capability to query their notification events via a REST API. This API has to support these capabilities:

- Query for all the event notifications per client allowing to filter by event creation date and delivery_status criteria.
- Obtain a single event notification details.
- Re-send a notification when delivery has definitely failed.

Candidate tasks

Task 1: System Design

- Design the overall solution to support both challenges, the delivery of event notifications as well as the self service API, as they are described above, ensuring scalability and resiliency.

Task 2: API implementation

- Using hexagonal architecture, implement a mechanism to obtain the event notification and deliver it to the proper webhook using https to an specific URL (to be provided in the presentation date).
- Using Java Spring Boot implements the capability of self service via API supporting `GET /notification_events`, `GET /notification_events/{notification_event_id}` and `POST /notification_events/{notification_event_id}/replay`.
Use the list of event notifications attached file named: `notification_events.json`.

Task 3: Security

- The Cobre API could be consumed publicly through the internet, then identify at least three OWASP Top 10 vulnerabilities that could impact your API.
- Propose security measures to mitigate each one of your choices.

Additional information

- You are free to use any format to deliver your solution proposal, could be a document or presentation. Also, feel free to use any modeling framework to document your solution (UML, C4, etc.).
- Please be prepared to discuss any architectural decision and reasoning you have made.
- You will present your case solution to a panel of people, please consider to be ready for questions.
- About implementation tasks, please share a github repository. Last commit should be before delivery date.

- We promote AI usage as a compliment to enhance productivity, so we encourage you to use it in your implementation, then please document in detail (prompts, screenshots, etc.) in what use cases you have used.
- Enjoy!